

T VINAY KUMAR

Tirupati, Andhra Pradesh

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EXPERIENCE

Light & Wonder

09/2025 – Present

Associate Software Engineer

Bengaluru, India

- Developed a high-performance GUI using LVGL for device flash management, featuring reusable UI components, WebSocket communication, and support for component installation, log management, and system configuration reducing memory footprint by 40% and improving GUI rendering performance by 60%.
- Resolved critical defects in ArgOS through systematic debugging, log analysis and leveraged an AI-powered workflow integrating GitHub Copilot with Jira, logs, and codebase context to perform root-cause analysis and generate targeted fixes, reducing average bug resolution time by 60%.

Tech Stack: React, C++, Redux, Event-Driven Architecture, RabbitMQ, Docker, Middleware

Light & Wonder

03/2025 – 09/2025

Software Engineer Intern

Bengaluru, India

- Developed 15+ automated test scenarios covering game cycles, event validations, and meter flows, improving test coverage and early defect detection by 40%.
- Resolved critical build warnings and major bugs in the ArgOS codebase through extensive debugging, log analysis, and root-cause analysis, resulting in a 20% improvement in overall system stability and response handling.

Tech Stack: React, Redux, TypeScript, C++, CI/CD, OS, Debugging, Microservices, WebSockets

Amazon

09/2023 – 10/2023

Amazon ML Summer School - Apprenticeship

- Involved in a 6-week intensive apprenticeship program focused on Machine Learning, covering foundational concepts, algorithms and practical AI Models in real-world scenarios.

Tech Stack: ML, Deep Learning, Deep Neural Networks, Reinforcement Learning, GenAI, NLP, AI

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, C, SQL, JavaScript

Frameworks & Database : React.js, Next.js, Node.js, Express.js, Spring boot, MySQL, MongoDB

Tools/Technologies: Git, Perforce, CI/CD, Docker, Kubernetes, Linux, Ubuntu

Core CS Fundamentals: OOPS, DSA, OS, Computer Networks, DBMS, NLP, ML, Distributed Systems

PROJECTS

Evo-Health | 🌐

- Designed and implemented a Genetically Optimized Bayesian Network for early Metabolic Syndrome prediction with optimal feature selection, severity classification, and personalized diet, exercise, and yoga recommendations by risk level.
- Developed a scalable full-stack healthcare application using Node.js, Express.js, Spring Boot MongoDB, and REST APIs for clinical assessment, risk visualization, health tracking, and personalized report generation.
- Integrated an LLM-powered AI chat assistant with real-time user health context and recommendations for personalized conversational guidance.
- Containerized Application using Docker and deployed applications on Minikube-based Kubernetes

Tech Stack: React, Node.js, Express.js, Java, Spring boot, MongoDB, JWT, REST APIs, LLMs, Gemini API, Axios, Python, Bayesian Networks, Docker, Kubernetes(K8s)

EDUCATION

Amrita Vishwa Vidyapeetham, Coimbatore

09/2021 – 05/2025

B.Tech - Computer Science and Engineering - CGPA - 8.18/10

Coimbatore, Tamil Nadu

PUBLICATIONS

- Published a research paper indexed in IEEE and SCOPUS: "A Hybrid Feature Selection Model for Early Prediction of Metabolic Syndrome", ICSADL 2025.